

Walter Zoff

Automation, Data & AI · AI Engineer / Forward Deployed Engineer

Milan, Italy | walter1zoff@gmail.com | +39 338 864 9369 | linkedin.com/in/walterzoff | github.com/walterzoff | walterzoff.github.io

Automation & Control engineer (MSc, Politecnico di Milano, 110/110) working across data and AI. I build BI platforms and LLM-powered tools (RAG, MCP), with research roots in data-driven modeling and control. Comfortable taking a model from a notebook to a reliable system that ships. Targeting AI Engineer / Forward Deployed Engineer roles.

EXPERIENCE

Junior Consultant — Data & AI

Nov 2025 – Present

Target Reply (Reply Group) · Milan

- Designed and built Looker / LookML dashboards and a BigQuery analytical layer (SQL data modeling and query optimization) for a leading Italian banking group.
- Manipulated and validated data in Python (Pandas); supported ETL and data orchestration across GCP and Databricks.
- Internal AI initiative: built an LLM orchestrator that automates MicroStrategy → Qlik dashboard migration via a Qlik MCP server and the Qlik REST APIs, including parsing of MicroStrategy HTML exports.
- Reply hackathon: built a RAG-based Change Impact Analyzer (transformers + LLM APIs, prompt engineering) for change-request effort estimation and data lineage, with a React / Node.js frontend.
- Led requirements gathering (metrics, KPIs, dashboard design) with stakeholders.

Instrumentation & Automation Engineer

Jun – Nov 2025

SIMECO S.p.A. · Milan

- Mechanized P&IDs and defined automation typicals (motors, control / on-off valves, safety devices) for industrial plant projects.
- Prepared instrument and I/O lists and technical specifications; supported procurement, FAT/SAT and field activities.

EDUCATION

M.Sc. Automation & Control Engineering — 110/110

2022 – 2025

Politecnico di Milano

- Thesis: trained a neural network (MLP) as a surrogate model of an industrial thermoforming process, used in an optimization loop to improve energy efficiency. Coursework: machine learning, optimization, MPC, system identification.

B.Sc. Automation Engineering — 107/110

2019 – 2022

Politecnico di Milano

PUBLICATIONS

- W. Zoff et al., "Data-Driven Control of Thermoforming Machines," *IEEE Transactions on Control Systems Technology*, 2026.
- W. Zoff et al., "Data-Driven Modeling and Optimization of the Thermoforming Heating Phase," *IFAC-PapersOnLine*, 2025.

SELECTED PROJECTS

- **LLM Orchestrator for BI Migration** (2026) — MicroStrategy → Qlik via LLM orchestration. Python, LLM APIs, MCP, Qlik APIs, HTML parsing.
- **Change Impact Analyzer** (2026) — full-stack RAG app for change-request effort and data lineage. RAG, transformers, LLM APIs, React, Node.js, SQLite.

- **Neural-Network Process Optimization** (M.Sc. thesis, 2025) — MLP surrogate in an optimization loop.
Python, PyTorch, optimization.

SKILLS

Languages	Python, SQL, C/C++, MATLAB, JavaScript
Generative AI	LLM APIs, RAG, prompt engineering, transformers, LLM orchestration, MCP
Machine Learning	PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, optimization
Data & BI	BigQuery, Databricks, GCP, Looker / LookML, Qlik Sense, MicroStrategy, ETL, data modeling
Control & Automation	control systems, MATLAB / Simulink, MPC, system identification, instrumentation, P&ID
Web & Tools	React, Node.js, SQLite, Git, FastAPI, Excel / VBA

CERTIFICATIONS & LANGUAGES

- | | |
|--|--------------------|
| • Google — Looker Skills | • Italian — native |
| • SAS Hackathon Boot Camp (2026) | • English — C1 |
| • MATLAB / Simulink for Control Systems (PoliMi) | • Spanish — A2 |